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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2023
END SEMESTER EXAMINATION - ODD 2023
PH(IT)101 - Engineering Physics - I

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. A hologram is a (1) CO1 BL1
 - a) 3D image of the object
 - b) 2D image of the object
 - c) 1D image of the object
 - d) None of these

2. Logic gate that is used in Pen drive (1) CO4 BL2
 - a) NAND
 - b) NOR
 - c) AND
 - d) XOR

3. How does a semiconductor behave at absolute zero? (1) CO2 BL2
 - a) Conductor
 - b) Insulator
 - c) Semiconductor
 - d) None of them

4. The Miller indices of a plane having intercepts of 8a, 4b and 2c on the a, b c axes respectively will be (1) CO3 BL3
 - a) (112)
 - b) (234)
 - c) (124)
 - d) (211)

5. In fiber optics, total internal reflection occurs at the _____. (1) CO1 BL2
 - a) Core-cladding interface
 - b) Core-air interface
 - c) Cladding-air interface
 - d) Core-core interface

6. Population inversion in preparing laser beam can be achieved (1) CO1 BL2
 a) When one of the excited states is less populated than the ground state
 b) When one of the excited states is more populated than the ground state
 c) When the population of one excited state and the ground state are equal
 d) On the basis of none of the above conditions
7. The atomic diameter of a BCC crystal (if a is lattice parameter) is (1) CO4 BL4
 (a) a
 (b) $a/2$
 (c) $\sqrt{3}a/2$
 (d) $a/(4/\sqrt{2})$
8. In CRO which of the following is a part of the electron gun? (1) CO1 BL1
 a) Cathode
 b) Grid
 c) Accelerating anode
 d) All of the above
9. Life Time of the metastable State is (1) CO1 BL2
 a) 1ns
 b) 1 μ s
 c) 1s
 d) 1ms
10. "Quantum dot" is an example of (1) CO2 BL1
 a) 1D object
 b) 2D object
 c) 3D object
 d) 0D object
11. Which of the following materials show 2-D confinement? (1) CO5 BL4
 a) CNT
 b) Graphene
 c) Graphite
 d) Fullerene
12. The ratio of de Broglie wave length of a moving proton and moving electron is (1) CO4 BL4
 a) 2:1
 b) 1:2
 c) 1:3
 d) 3:1

Group B

(Short Answer Type Questions)

(Answer any three of the following) 3x5=15

- 13a. Distinguish between step-index & graded-index fiber? (2) CO1 BL3

- 13b. Find Compton wavelength for an electron. (3) CO5 BL2
- 14a. Calculate the kinetic energy of a neutron having de Broglie wavelength 5×10^{-16} m. The rest mass of the neutron is 1.675×10^{-27} Kg. (2) CO5 BL3
- 14b. Silicon crystallizes in the diamond cubic structure. The radius of the silicon atom is 0.1176nm. The atomic weight and density of silicon are 28.09 and 2.3×10^3 Kg/m³. What is the number of atoms present in a unit cell? (3) CO3 BL3
15. Write short notes on: (5) CO4 BL1
- a. Carbon nanotube
 - b. Graphene
 - c. Quantum dot
 - d. LED
 - e. LCD
- 16a. Compare de Broglie wavelength of car of mass 6000 kg moving with velocity 30 m/second with an electron of mass 9.31×10^{-31} kg moving with speed 10^7 m/sec. (3) CO3 BL3
- 16b. Give examples of intrinsic and extrinsic semiconductors (two in each case). (2) CO2 BL1
- 17a. Find out the atomic diameter for a BCC structure. (3) CO1 BL5
- 17b. Consider a single photon as matter with energy $E = hu$ and momentum $p = hu/c$ then if it shows wave nature what would be expression for its wave length? (2) CO3 BL5

Group C

(Long Answer Type Questions)

(Answer any three of the following) 3x15=45

- 18a. Find the ratio of the rate of spontaneous emission to the rate of stimulated emissions at $T = 1000$ K for i) visible radiation of frequency 5×10^{14} Hz and ii) microwave radiation of frequency 10^9 Hz . Comment on the results. (5) CO1 BL4
- 18b. What do you mean by population inversion related to laser? Illustrate with a necessary diagram. Is M-B distribution law is obeyed here? Explain. (3) CO1 BL4
- 18c. Deriving Einstein's equation for LASER transition, show that the ratio between spontaneous and stimulated emission is proportional to the cube of frequency. What are the physical interpretations of Einstein's equations? (7) CO1 BL4
- 19a. Obtain expressions of Acceptance Angle and Numerical Aperture in relation to the working of an optical fiber. Hence establish a relation between them. (7) CO1 BL2

- 19b. Describe different components of CRT with proper diagram. (6) CO4 BL3
- 19c. What is the physical significance of wave function. (2) CO5 BL4
20. Answer the following: (15)
- a) Explain the operation and application of CRT and CRO. (5) CO4 BL2
- b) What is Nanomaterial? Explain Quantum Wells, Quantum Wires, and Quantum Dots. (5) CO2 BL2
- c) i. What do you mean by acceptance angle in an optical fibre. (5) CO1 BL4
 ii. State the differences between Step-Index and Graded-Index optical fibre.
 iii. A step-index optical fibre has refractive indices of the core and cladding 1.44 and 1.40, respectively, and is surrounded by air. Calculate the numerical aperture.
- 21a. When a radiation of frequency is incident on a metal surface, electrons are emitted with maximum energy. What is the lowest frequency of radiation required for emission of electrons from the metal surface? (5) CO5 BL3
- 21b. Derive the relativistic expression of matter wave in terms of rest mass energy and kinetic energy. (6) CO5 BL4
- 21c. Estimate wavelength of electron wave based on the observations of Davission germer experiment made on $V_1=54$ V and scattering angle= 50 degree. (4) CO3 BL5
22. Answer: (15)
- a) Find the atomic packing factor for SC, FCC and BCC lattices. (5) CO3 BL3
- b) State Heisenberg's uncertainty principle. Calculate the wavelength associated with an electron carrying energy 2MeV. (5) CO5 BL3
- c) Differentiate between conductor, semiconductor, and insulator based on band diagram.. (5) CO3 BL3
 An insulator has an optical absorption that occurs for all wavelengths shorter than $1800\text{\AA}$. Find the width of the forbidden energy band for this insulator.